



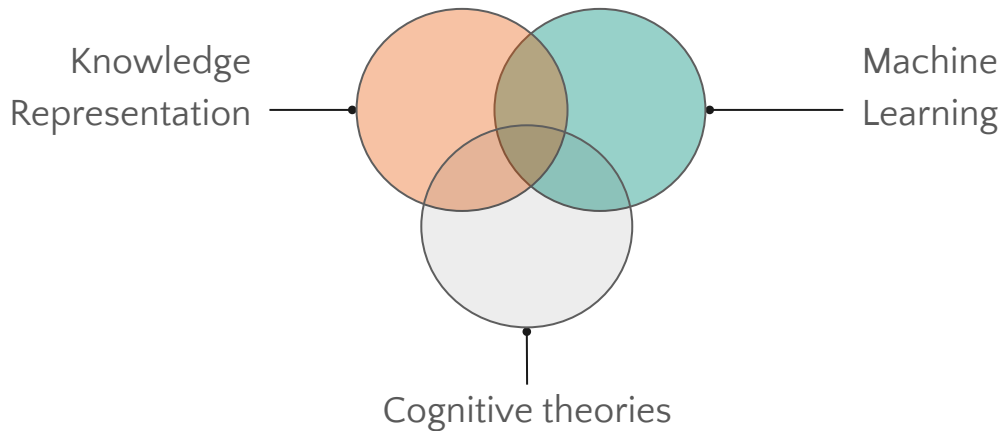
Hybrid Intelligence

The next AI generation



AI that **collaborates** with people, instead of replacing them

A bit of myself



eScience

AI to support scientists
and their questions

Robotics

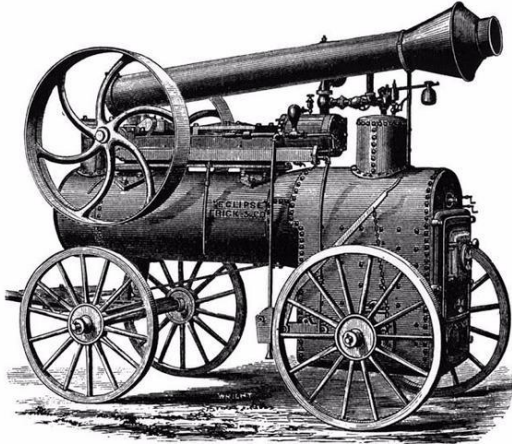
Facilitating tasks of
mobile agents with AI

Ethical AI

Ethical methods to design
intelligent systems

Let's take a few steps back

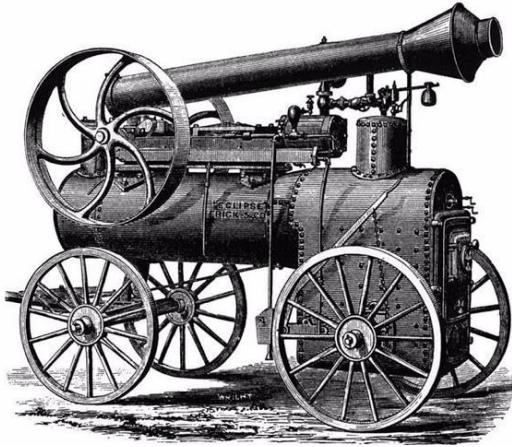
Science helped **extending** human capabilities



steam engine = **muscle power**

Let's take a few steps back

Science helped **extending** human capabilities



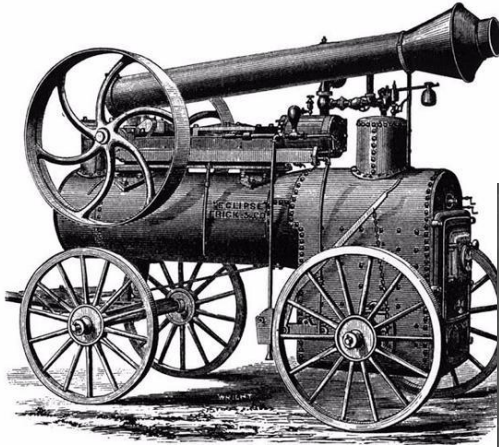
steam engine = **muscle power**



car = **human speed**

Let's take a few steps back

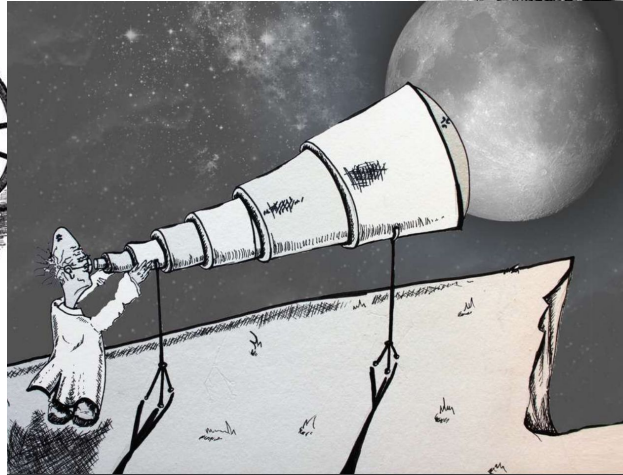
Science helped **extending** human capabilities



steam engine = **muscle power**



car = **human speed**



telescopes = **human vision**



Let's take a few steps back

Artificial Intelligence will help extending **human intelligence**

AI



Let's take a few steps back

Do we have a definition for “intelligence”?

A?

Classical AI definition : the Turing test

How to **measure** a machine's intelligence:

- a human judge
- a machine VS a human conversing in natural language
- all separated

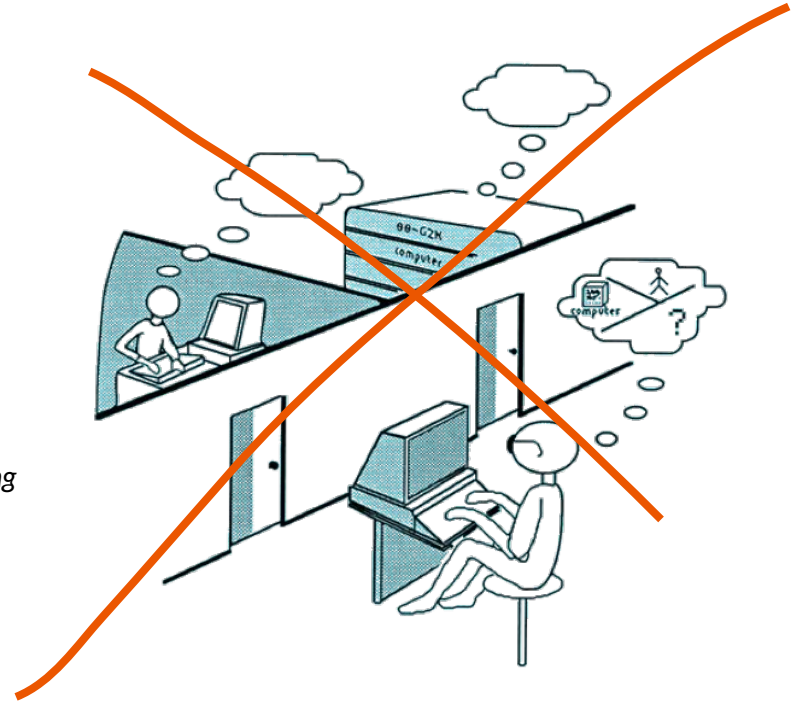
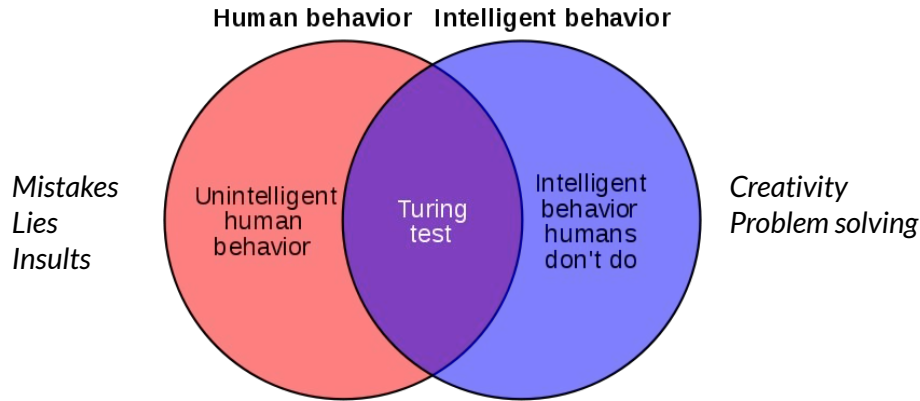
If the judge cannot tell the machine from the human, the machine **passes the test**



Classical AI definition : the Turing test

No definition of intelligence!

- only how to recognise it
- only if the machine behaves as a human
- too anthropocentric



AI should not imitate human intelligence



Imagine if planes were imitating birds.

AI should not replace human intelligence



It'd be like fitting the wrong nail

AI should collaborate with human intelligence



There comes Hybrid Intelligence!



Hybrid Intelligence

Where to start?

4 functionalities

Collaborative

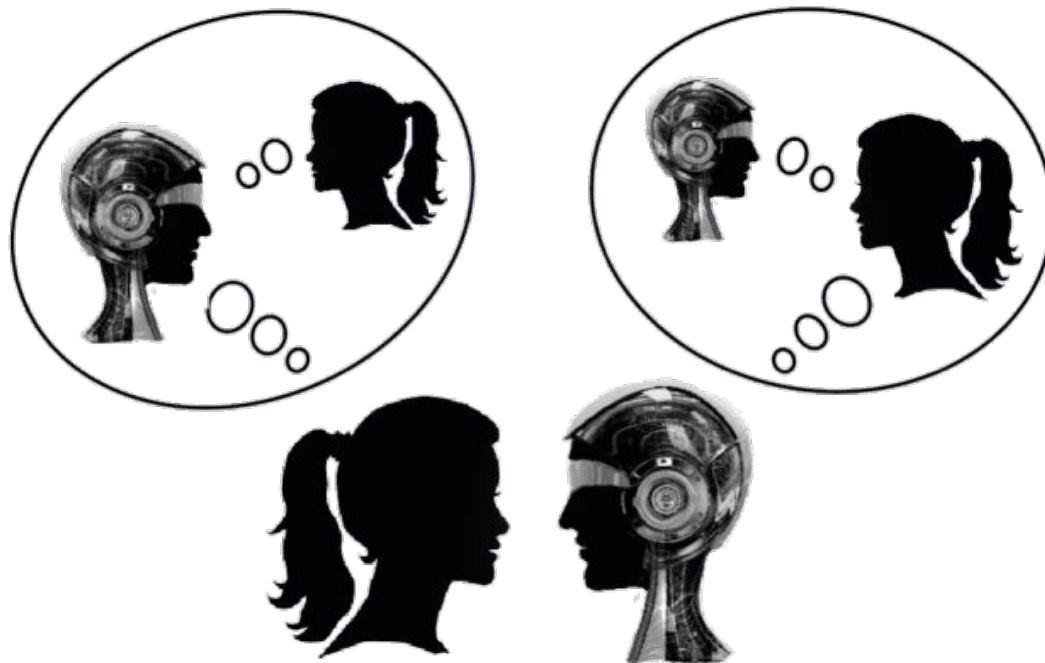
Adaptive

Responsible

Explainable

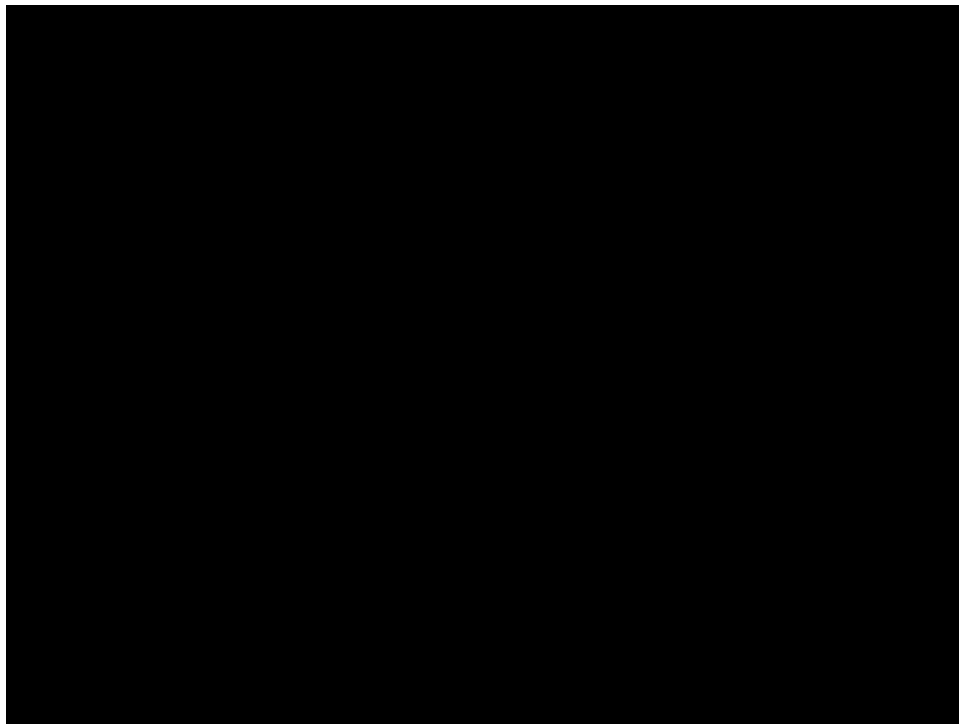
1. Collaborative HI

Humans do mindreading all the time





1. Collaborative HI

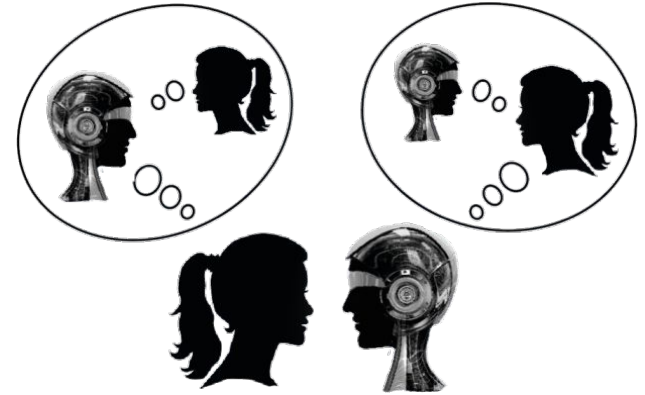


1. Collaborative HI

What is ToM?

Theory of mind

“The ability to attribute a wide range of mental states to the others, including intentions, hopes, expectations, imaginings, desires and beliefs, and thereby interpret, explain, and predict their behaviour.”

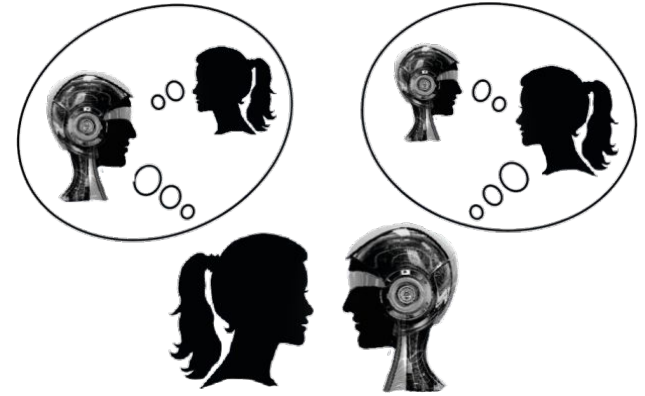


1. Collaborative HI

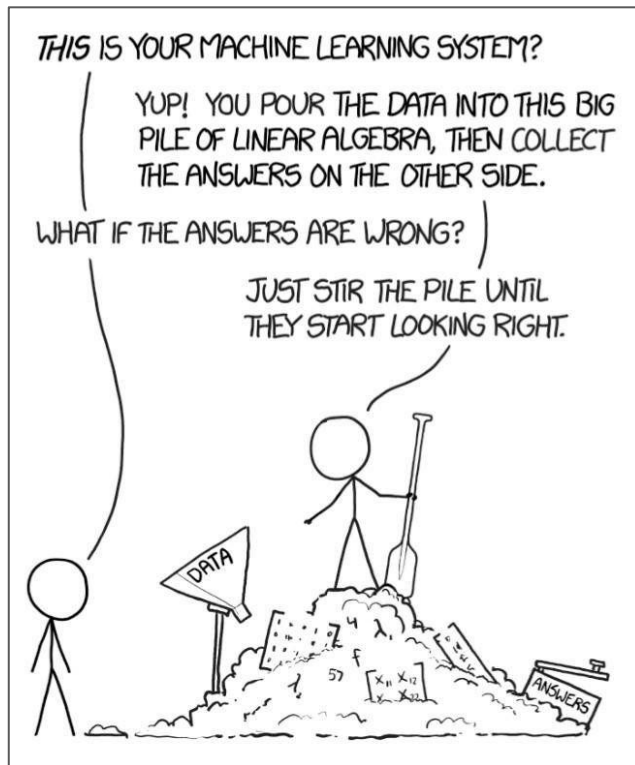
What is ToM?

Theory of mind

“The ability to attribute a wide range of mental states to the others, including intentions, hopes, expectations, imaginings, desires and beliefs, and thereby interpret, explain, and predict their behaviour.”



2. Adaptive HI

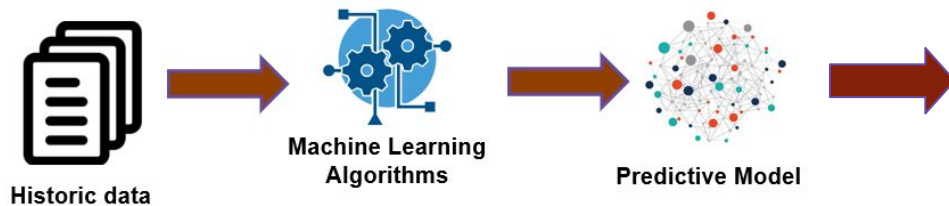


Current state of Machine Learning.

2. Adaptive HI

A primer on Machine Learning

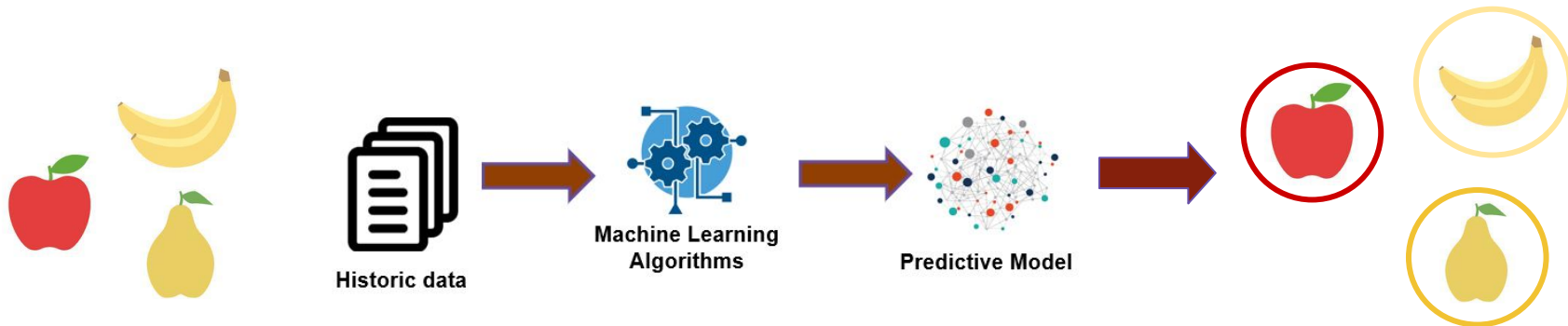
- Learn on historical data (**Training**)



2. Adaptive HI

A primer on Machine Learning

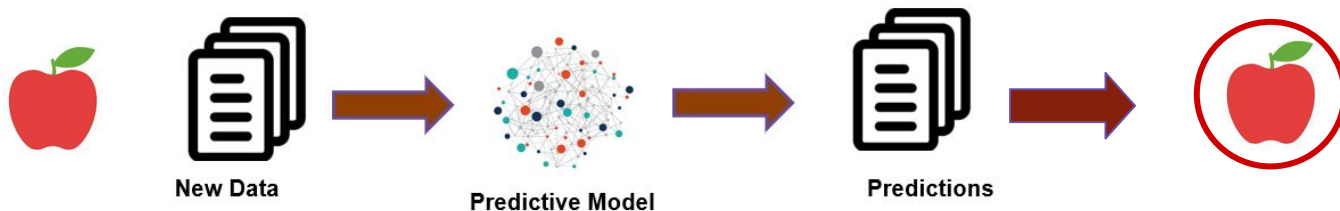
- Learn on historical data (**Training**)



2. Adaptive HI

A primer on Machine Learning

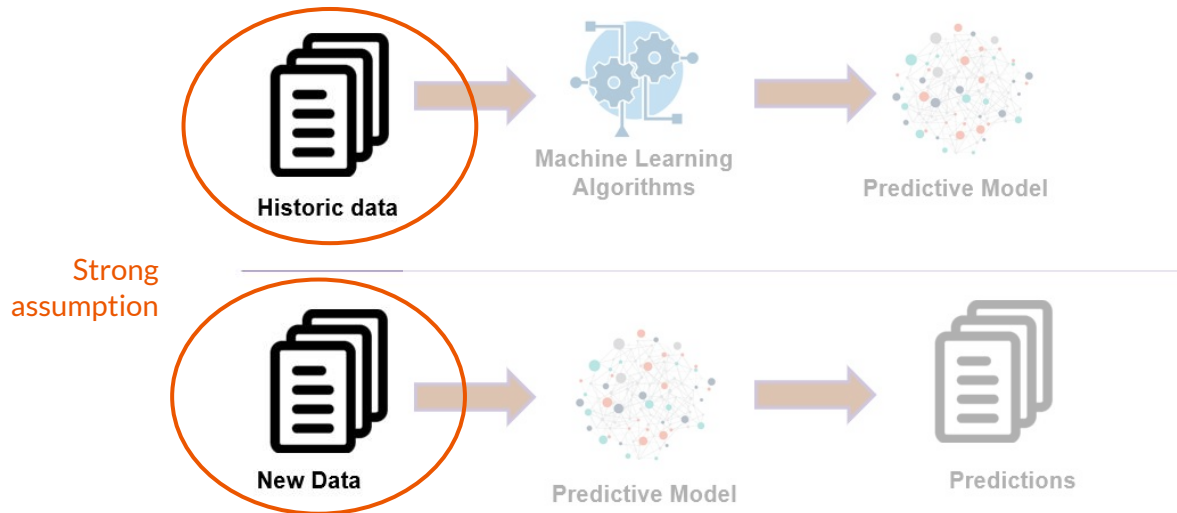
- Learn on historical data (**Training**)
- Apply over new data (**Predicting**)



2. Adaptive HI

A primer on Machine Learning

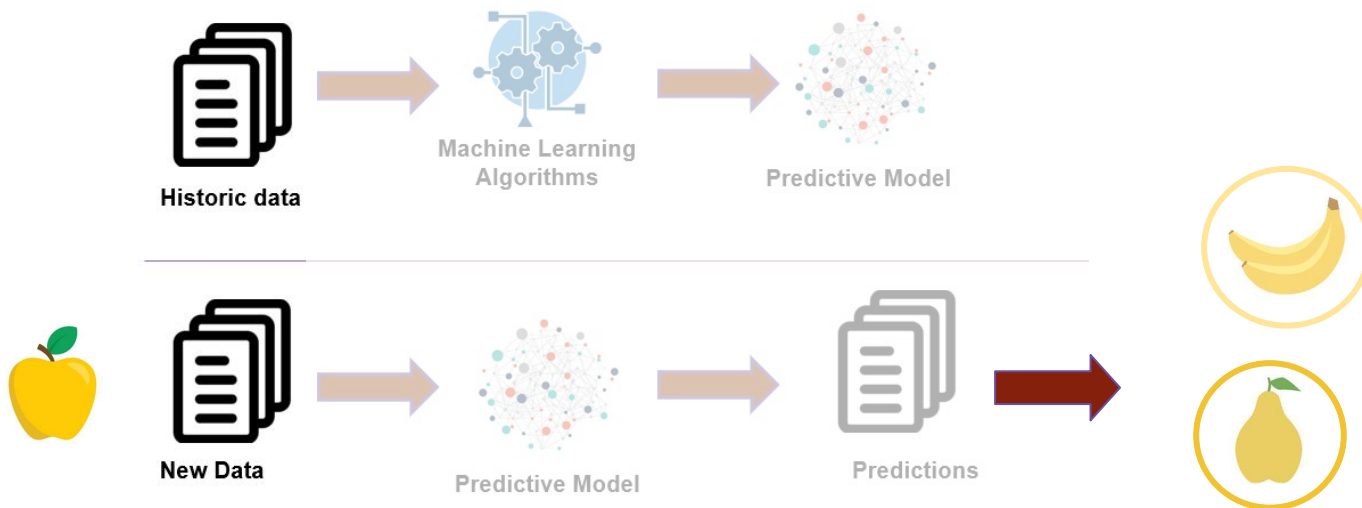
- Learn on historical data (**Training**)
- Apply over new data (**Predicting**)



2. Adaptive HI

A primer on Machine Learning

- Learn on historical data (**Training**)
- Apply over new data (**Predicting**)





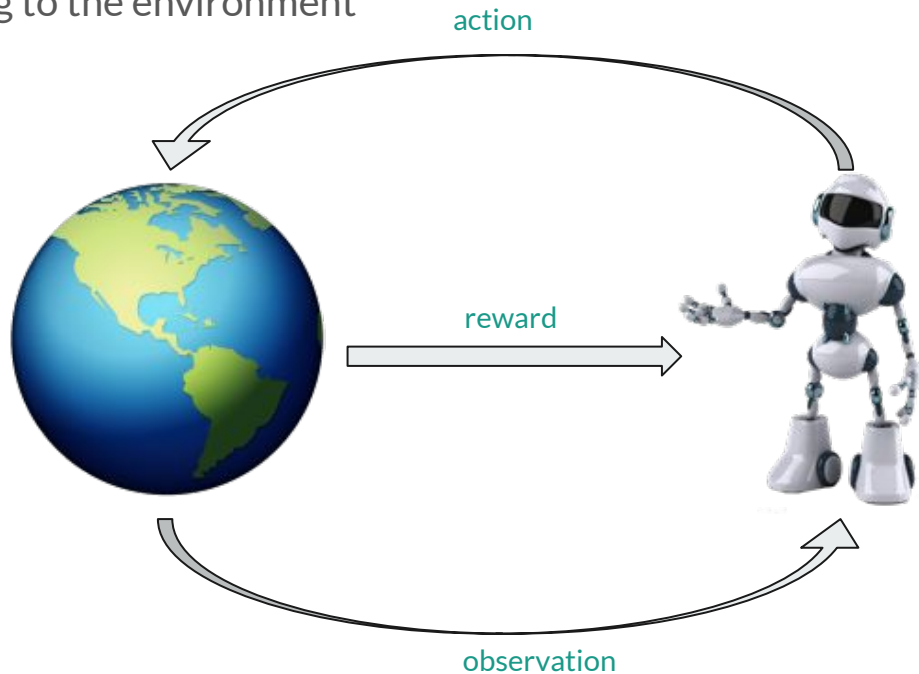
Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

2. Adaptive HI

Reinforcement Learning for the rescue

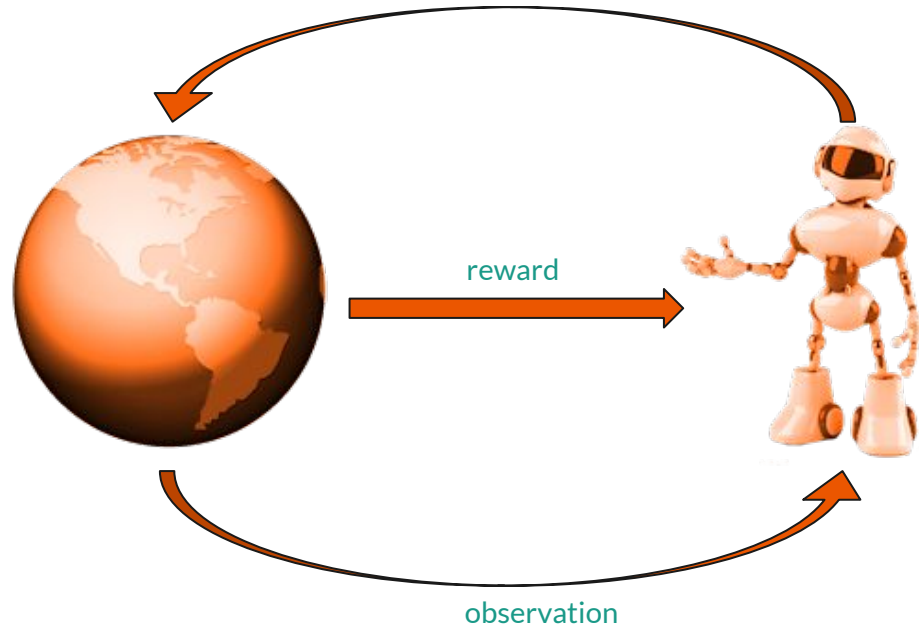
- i.e. adapting to the environment



2. Adaptive HI

Reinforcement Learning for the rescue

- rewards&actions in different worlds action



3. Responsible HI

The classical approach : **implicit** human values



3. Responsible HI

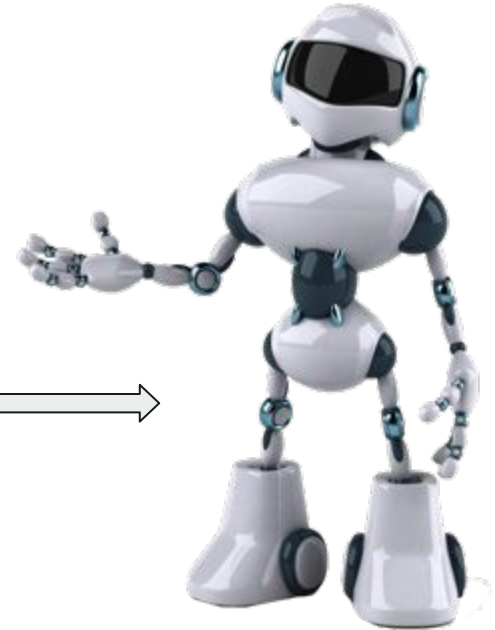
A new approach : ethics **by design**



values
ethics
norms



actions

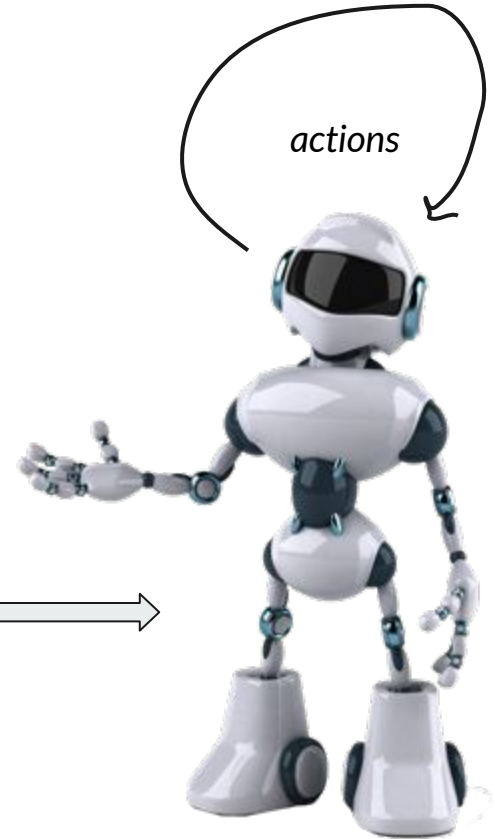


3. Responsible HI

A new approach : ethics **by design**



values
ethics
norms



4. Explainable HI

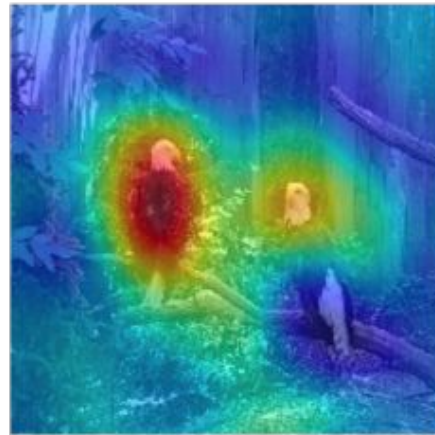
Current approaches : data-driven



(d) Bird - 100%, Person - 39%



(e) Importance map of 'bird'

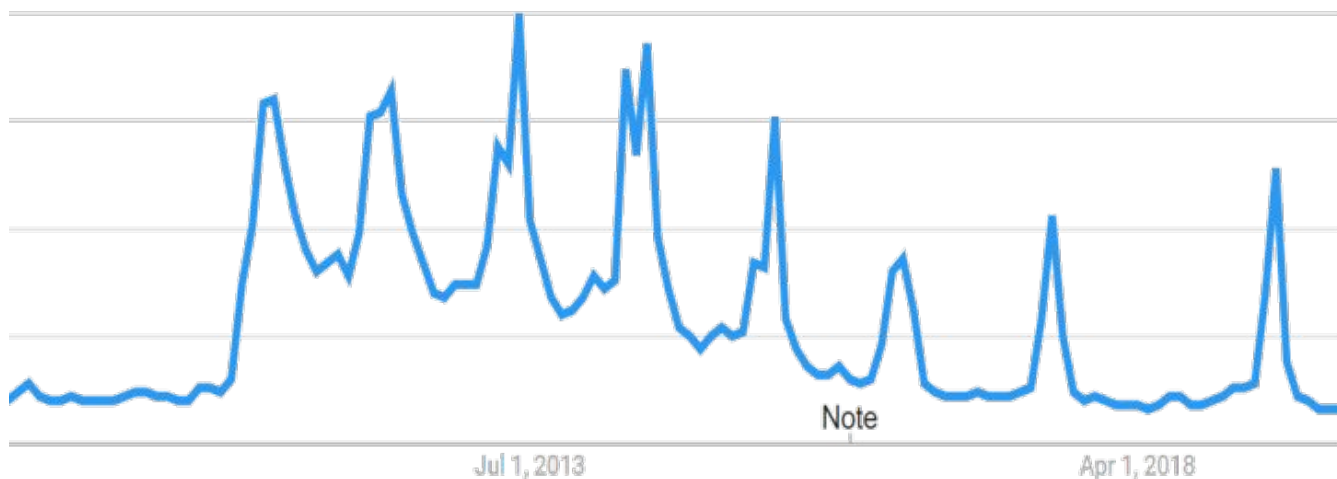


(f) Importance map of 'person'

Only developers can really understand this explanation. What about users?

4. Explainable HI

Humans explain with **background knowledge**



4. Explainable HI

Humans explain with **shared** background knowledge



Class: 793

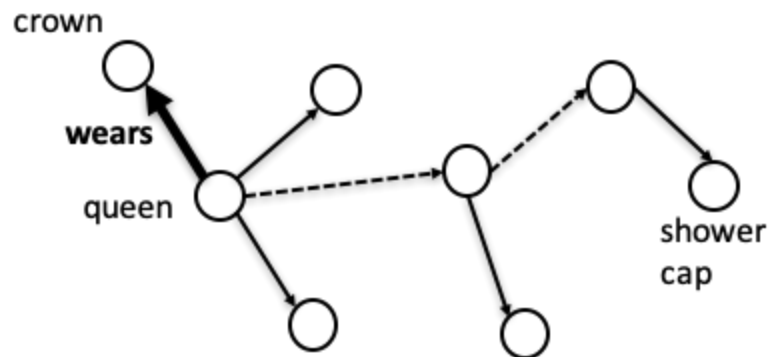
Label: n04209133

(shower cap)

Certainty: 99.7%

4. Explainable HI

New approach : **knowledge graphs** as background knowledge to **explain data**





A recap of Hybrid Intelligence

Hybrid Intelligence

- AI to **enhance** human capabilities (as others tools do)
- **Collaborate** with humans instead of replacing them

CARE : 4 HI lines

1. Collaborative HI → from a theory about the world to **the others' theories** of the world
2. Adaptive HI → from Machine Learning to **Reinforcement Learning**
3. Responsible HI → from implicitly to **explicitly coded** norms&values
4. Explainable HI → from data-driven to **hybrid explanations** (data&knowledge-driven)



Questions?

All slide credits to Frank van Harmelen and the HI team

First, a break

See you at 11:00!



Hybrid Intelligence: applications and challenges



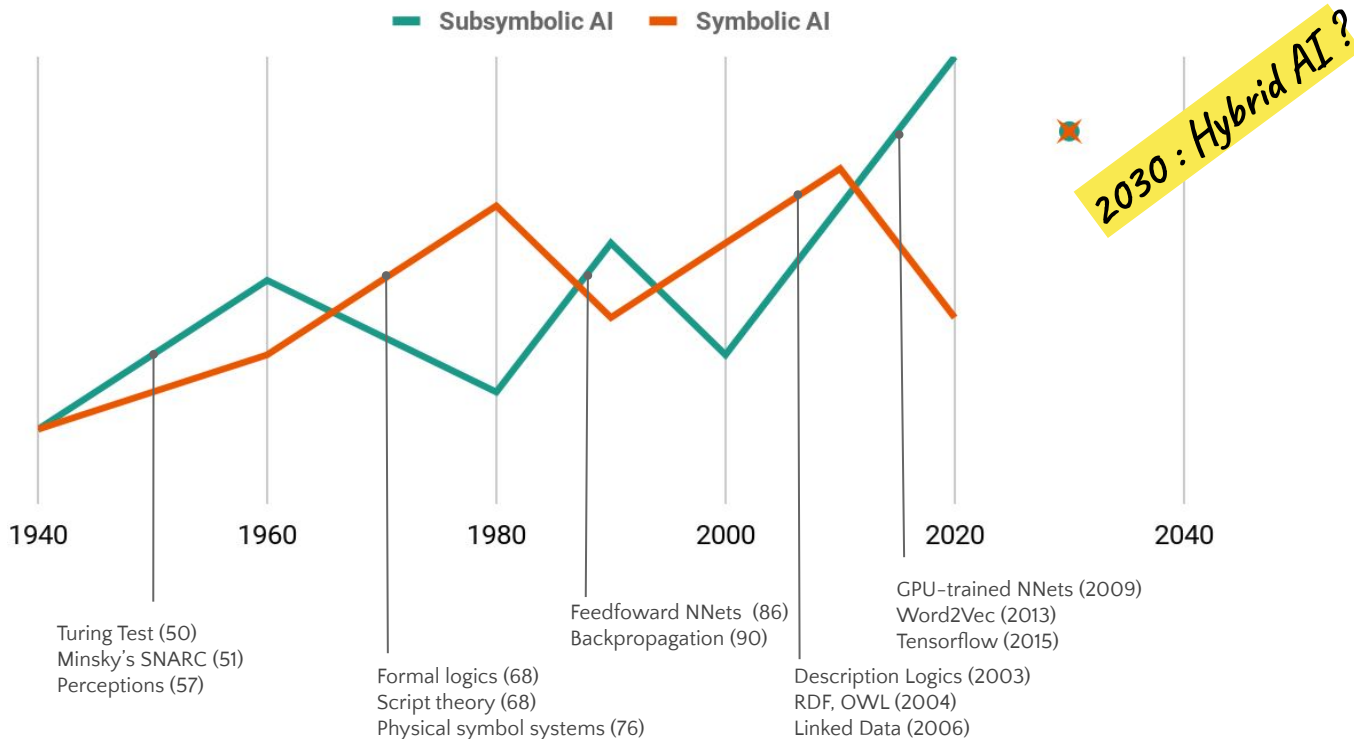
How far has HI gone already?

Let's dig from theory to applications:

1. Combining symbolic (Knowledge Representation) and subsymbolic methods
(Machine Learning)
2. Integrating humans and AI capabilities
3. Explicit integration of norms&values

Combining symbolic and subsymbolic methods

A (very simplified) history of AI



Dedalo : Explaining with Knowledge Graphs

How does it work?

- Take any **ML output** data



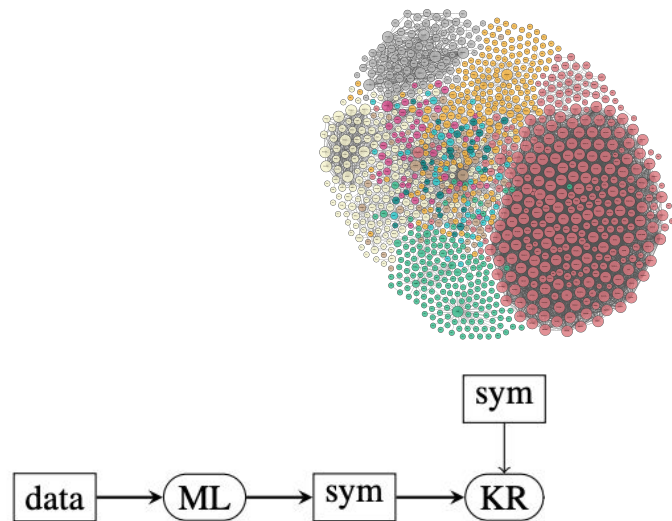
Tiddi, I. et al. (2014). *Walking Linked Data: a graph traversal approach to explain clusters*. COLD @ ISWC 2014.

van Harmelen et al (2019). [*A Boxology of Design Patterns for Hybrid Learning and Reasoning Systems*](#). Journal of Web Engineering (2019).

Dedalo : Explaining with Knowledge Graphs

How does it work?

- Take any **ML output** data
- Use **KR techniques** knowledge about such data (triples in knowledge graphs)



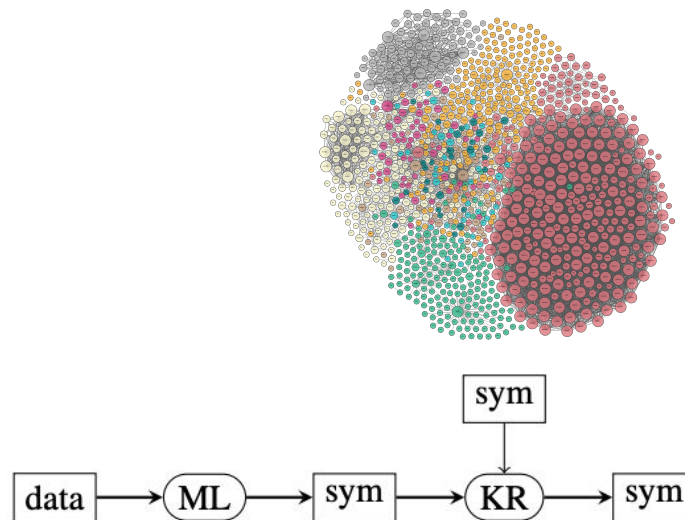
Tiddi, I. et al. (2014). *Walking Linked Data: a graph traversal approach to explain clusters*. COLD @ ISWC 2014.

van Harmelen et al (2019). [A Boxology of Design Patterns for Hybrid Learning and Reasoning Systems](#). Journal of Web Engineering (2019).

Dedalo : Explaining with Knowledge Graphs

How does it work?

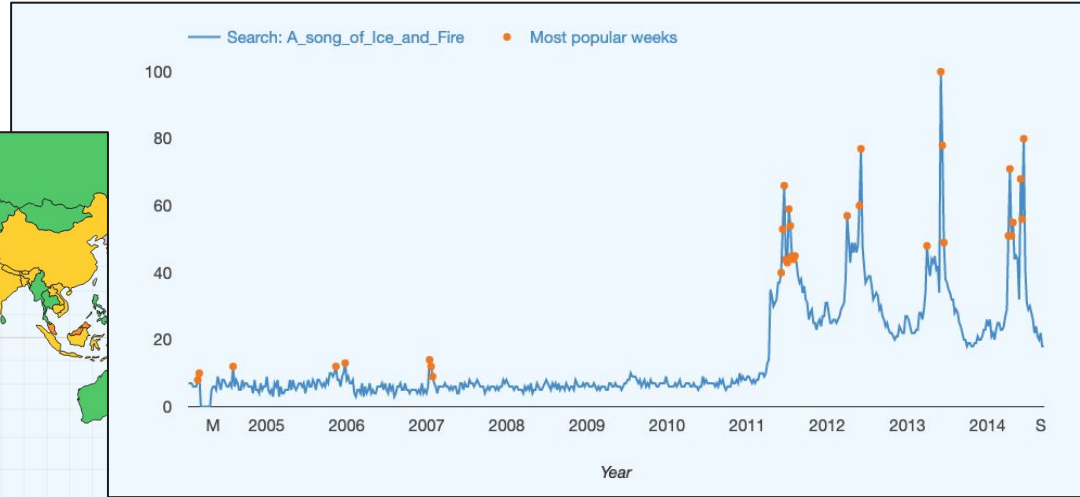
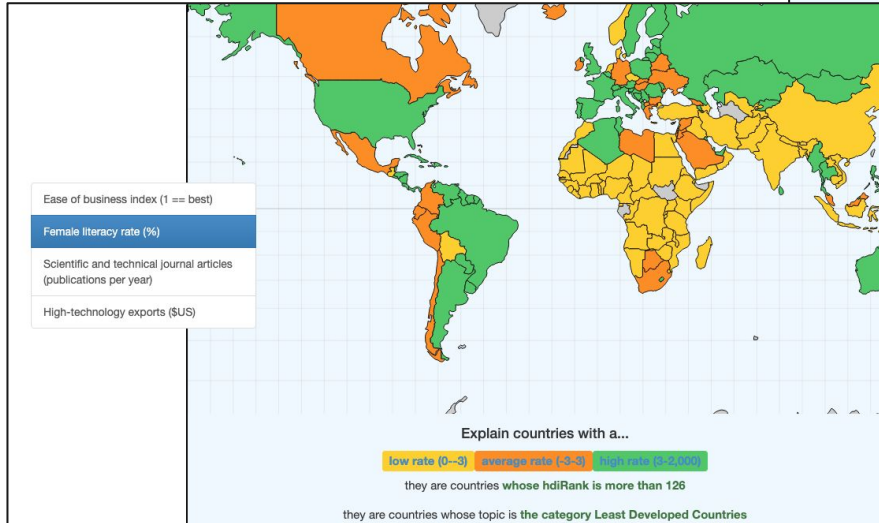
- Take any **ML output** data
- Use **KR techniques** knowledge about such data (triples in knowledge graphs)
- Use the knowledge to **derive explanations** about subsets of data



Tiddi, I. et al. (2014). *Walking Linked Data: a graph traversal approach to explain clusters*. COLD @ ISWC 2014.

van Harmelen et al (2019). [A Boxology of Design Patterns for Hybrid Learning and Reasoning Systems](#). Journal of Web Engineering (2019).

Dedalo : Explaining with Knowledge Graphs



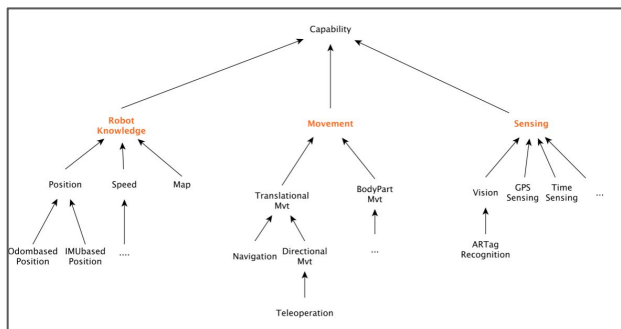
*People search for "A song of Ice and Fire"
because a new season of the
Game of Thrones TV series is out*

*Countries have a female low literacy rate
because they are less developed countries*

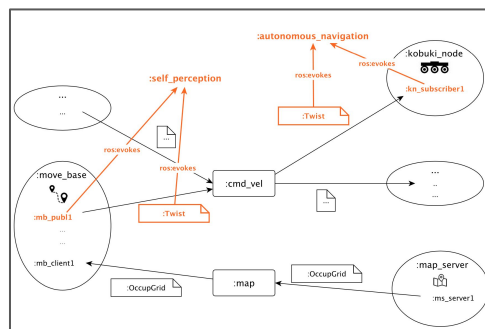
Abstracting robotic tasks with ontologies

Robots are **data hungry** (labelled data), highly task specific, but **symbolic AI** can help

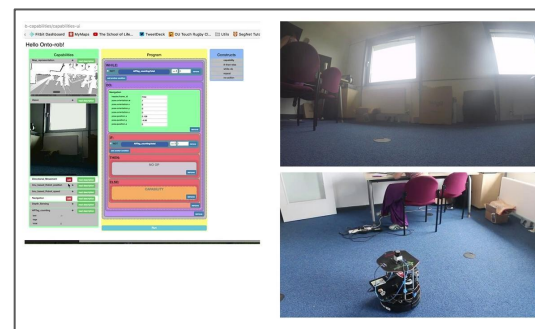
Taxonomy of high-level capabilities



Map robot components and capabilities in a KG



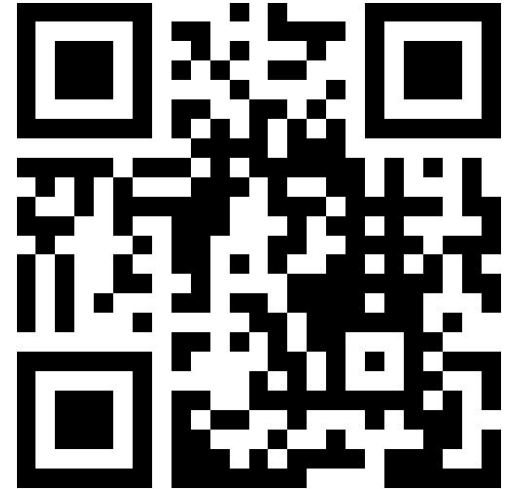
Allow reasoning on the capabilities of any robot



Mentimeter time

<https://www.menti.com/siacubwhiu>

Code : 20 60 63 8



Integrating humans and AI capabilities

Symbolic Autonomy

- Awareness of their perceptual, physical, and reasoning **limitations**
- Proactive **request for help** from humans, e.g. object manipulation actions



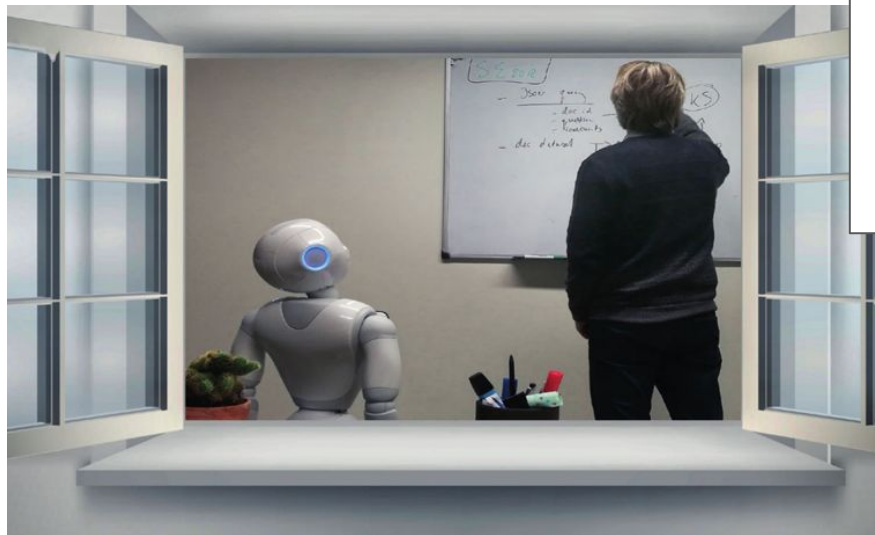
Extending AI capabilities

Robots accompanying teachers:

- Repetitive tasks
- Patience with mistakes
- Personalisation



The Al.nstein



Can a Transformer Assist in Scientific Writing? Generating Semantic Web Paper Snippets with GPT-2

Albert Meroño-Peñuela¹, Dayana Spagnuolo¹, and GPT-2²

¹ Vrije Universiteit, Amsterdam, The Netherlands

albert.merono@vu.nl, d.spagnuolo@vu.nl

² OpenAI Inc., San Francisco, California, US

Abstract. The Semantic Web community has produced a large body of literature that is becoming increasingly difficult to manage, browse, and use. Recent work on attention-based, sequence-to-sequence Transformer

Discovering Hypotheses using Knowledge Graph Embedding: The Case of Social Science

Rosaline E. de Haan, MSc

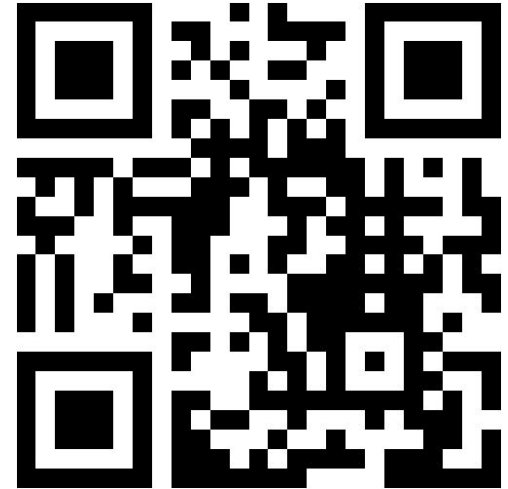
Gil, Y. 2017. *Thoughtful Artificial Intelligence*, Y. Gil, 2017.

Meroño-Peñuela A. et al, *Is a Transformer Your Next Semantic Co-Author? Generating Semantic Web Paper Snippets with GPT-2*, ESWC2020, demo : <http://swgpt2.amp.lod.labs.vu.nl>

Mentimeter time

<https://www.menti.com/siacubwhiu>

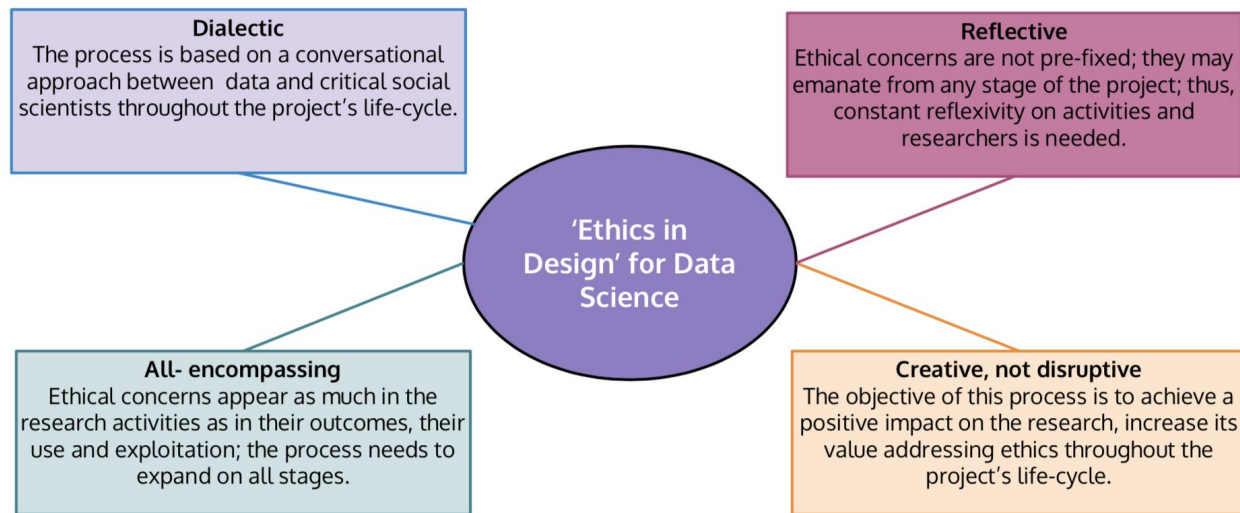
Code : 20 60 63 8



Explicit integration of norms&values

Ethics in design for Data Science

- Conversation between social and data scientists
- Throughout activities
- Constant update
- Positive outcomes



<https://kmitd.github.io/recoding-black-mirror/>

d'Aquin et al, *Towards an "Ethics in Design" methodology for AI research projects*, in AIES 2018.

Pro-social Machines

- Sociality
 - teamplayer
 - friendliness
- Empathy
 - Emotions
 - Self-confidence
- Pedagogy

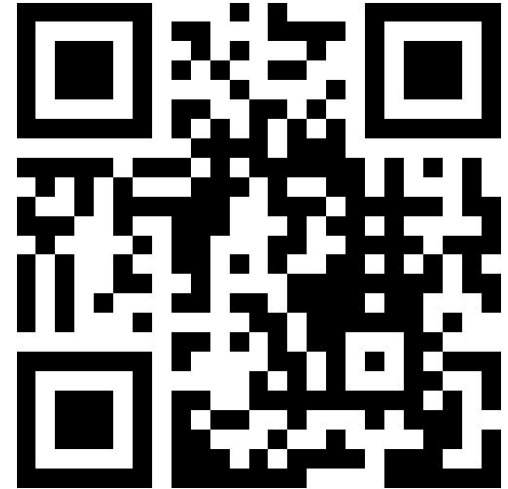


Paiva et al. (2020), [Towards prosocial design: A scoping review of the use of robots and virtual agents to trigger prosocial behaviour](#). Computer in Human Behaviors.

Mentimeter time

<https://www.menti.com/siacubwhiu>

Code : 20 60 63 8





Which HI research lines?

1. Combining symbolic (Knowledge Representation) and subsymbolic methods (Machine Learning) → Explainable HI, Adaptive HI
2. Integrating humans and AI capabilities → Collaborative HI, Responsible HI
3. Explicit integration of norms&values → Responsible, Explainable, Adaptive HI

An exercise

1. Think of a **technology** of these days (Alexa, Robot assistants, Virtual Reality, the ones seen in this lecture...).
2. What are its current **limitations** w.r.t. the HI research lines?
3. How could it be **extended** to integrate one (or more) line?

<https://www.menti.com/siacubwhiu> code 20 60 63 8



Results : <https://www.mentimeter.com/s/8d0a74223709db2a3d215a83b0309c54/761137e22614>



Thanks

Questions?

Hybrid Intelligence 2020-2030

Aimee van Wylsberghe

Annette ten Teije

Antske Fokkens

Bart Verheij

Birna van Riemsdijk

Catholijn Jonker

Christof Monz

Dan Balliet

Davide Grossi

Eliseo Ferrante

Florian Kunneman

Frank Dignum

Frank van Harmelen

Frans Oliehoek

Gusztai Eiben

Hayley Hung

Henry Prakken

Herke van Hoof

Holger Hoos

Jakub Tomczak

Koen Hindriks

Maarten de Rijke

Mark Neerincx

Max Welling

Myrthe Tielman

Piek Vossen

Rineke Verbrugge

Roel Dobbe

Silja Rennooij

Stefan Schlobach

Victor de Boer

Virginia Dignum

3. Virtual assistants with norms and values

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